

Assessing Atmospheric Corrosivity in Antarctica

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1. INTRODUCTION

It has generally been believed that corrosion rates are very low in the Polar regions because of the extreme coldness and dryness of the climate. In 1960 when snow and ice were removed from Scott's hut the contents were generally found to be in excellent condition. However, the unnecessary removal of the snow and ice has radically changed the microenvironment causing serious corrosion problems. Tins of foodstuffs which exhibited bright clean labels at the time the hut was cleared are now rust-stained and disintegrating¹.

The Antarctic Heritage Trust and the Antarctic Division of the New Zealand Department of Scientific and Industrial Research have a continuing management program for the three historic huts built by Shackleton and Scott on Ross Island. At both Shackleton's hut at Cape Royds (1908) and Scott's hut at Cape Evans (1911) there is a considerable amount of equipment and canned provisions outside the huts. Corrosion of this historic material is a matter of concern. The environment at these sites would appear to be severely corrosive and erosive with very strong winds frequently blowing salt-laden snow off the sea-ice and depositing it on the huts and equipment. The canned provisions were stockpiled in iron-framed plywood cases which are disintegrating and exposing the contents to salt-water attack.

As part of the program to preserve these historic sites, measurements have been made of the external levels of atmospheric corrosivity at both a coastal and an inland site in Antarctica.

2. EXPERIMENTAL

2.1 Exposure Sites

The coastal site was at Scott's Hut, Cape Evans, on Ross Island and is depicted in Figure 1 with Mt Erebus (3794m) clearly visible in the background. The inland site was at Vanda Station in the Wright Valley on the Antarctic continent, on the Western side of McMurdo sound, opposite Ross Island. The site is shown in Figure 2, the specimen being fixed to an antenna mast as indicated in Figure 3. This site is located in what is known as a "dry valley", one of the few regions in Antarctica that is virtually snow- and ice-free². Details of both sites are given in Tables 1 and 2.

Table 1. Details of exposure sites in Antarctica

Site	Latitude	Longitude	Location	Distance to coast km	Elevation m
Scott's Hut, Cape Evans	77° 38'S	166° 24'E	Coast	0.02	5
Vanda	77° 31'S	161° 40'E	Inland	44	130

2.2 Receptor Specimens

Specimens of standard low-alloy copper steels (100 × 50 × 3.14 mm) as used in previous studies were chosen^{4,5}. At Cape Evans two specimens of BHP Steel (C 0.16, Mn 0.61, Si 0.05, P 0.01, S 0.026, Ni 0.27, Cr 0.12, Cu 0.22%) and one specimen of BISRA steel (C 0.18, Mn 0.59, Si 0.031, P 0.01, S 0.034, Ni 0.27, Cr 0.12, Mo 0.04, Cu 0.22%) were exposed. At Vanda Station only one specimen of the BHP steel was exposed. The specimen pretreatment was as described previously⁴. At the time of recovery the specimens were taken to Scott Base, oven-dried at 110°C for 9 hours, sealed in plastic bags and despatched to New Zealand. The specimens were cleaned in Clarke's solution and the corrosion rates determined also as described previously⁴.

Five standard CLIMAT test specimens⁶ supplied by Alcan NZ Ltd were installed inside and outside at three historic hut sites in December 1982. Four of these were retrieved after exposure periods of only 41 or 42 days but showed no visible signs of corrosion and analysis was considered not worthwhile.

3. RESULTS AND DISCUSSION

Details of the dates of exposure and recovery and the calculated corrosion rates expressed in $\mu\text{m/y}$ are given in Table

Table 2. Climatic parameters for sites in Antarctica

Mean monthly temperatures, °C:		
	Scott Base (77°51'S, 166°45'E) (1957-1980)	Vanda Station (1968-1980)
January	-5	+1.2
February	-10.8	-5.9
March	-20.4	-20.4
April	-24.2	-29.7
May	-27.4	-29.2
June	-26.2	-30
July	-29.4	-38
August	-31	-32.3
September	-28.1	-31.2
October	-22.7	-15.7
November	-12	-6.5
December	-5.4	+0.3
Mean annual	-20.2	-19.8

Note on Precipitation:

The precipitation on Ross Island (Scott Base and Scott's Hut at Cape Evans) is 150-200 mm of water equivalent/year.

At Vanda station measurements of local snowfall are available only for three full years: 1969 — 82 mm, 1970 — 7 mm, and 1974 — 115mm snowfall³. Using a conversion factor of 1/16 to relate density of snow to that of water this represents a rain precipitation equivalent of between 0.5 and 7 mm/year.



Figure 1. Coastal site at Scott's Hut, Cape Evans, Ross Island with Mt. Erebus in the background. Photo by G.J. Dougherty.

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3. The one year corrosion rate at Cape Evans is substantial and about the same as that measured in Melbourne's outer north-eastern suburbs⁴. In that study there were 56 sites for which results were available for both BHP and BISRA steels, the respective mean results being 16 $\mu\text{m/y}$ and 15.2 $\mu\text{m/y}$ and the mean BHP/BISRA ratio was 1.058⁷. The ratio of the BHP and BISRA results at Cape Evans is virtually identical to this.

The reduction in the corrosion rate for three years exposure as opposed to one year is greater than expected. Results for the corrosion of a plain carbon steel exposed in all states of Australia in a range of environments showed 3 year corrosion rates to be about 60% of those for 1 year's exposure⁸. It is possible that corrosion products initially formed in the Antarctic may offer greater protection to the underlying metal thereby significantly reducing the rate of corrosion.

The result for Vanda station is extremely low. This site is located in one of Antarctica's most intriguing areas, the 'dry valleys'. This region was originally glacial but the glaciers have now receded leaving bare, deep valleys on the edge of the east Antarctic ice sheet. Cold, dry air from the ice sheet is warmed as it is compressed and descends up to 3000 metres to the valley floors as strong and extremely dry winds. These often exceed 100 kph, have a relative humidity of less than 10% and

evaporate any snow brought into the valleys by the moist coastal easterly winds². Local snowfall is very slight (see table 2), and rainfall **extremely** rare. On the occasions when rain was observed, the amount was barely sufficient to dampen the ground³. Incongruously, however, there is a permanently frozen freshwater lake at Vanda fed in summer by an inland flowing river sourced by melt water from glaciers at the eastern end of the Wright Valley⁹. With very little local snow, virtually no rain, and very dry winds, the valleys are the driest places on earth where temperatures range between -80 and $+15^{\circ}\text{C}$ ². This 'desert' environment is clearly apparent in Figure 2.

For comparison, the corrosion rates of carbon steel and ingot iron at locations in subpolar and hot arid climates are given in Table 4. They are all significantly lower than Cape Evans but only Norman Wells is

of the same order as Vanda. This Canadian site some 450 km inland has long been the benchmark for a site with low corrosivity. It is most surprising that the results at Norman Wells are as low as reported in view of its climatic parameters given in Table 5. Russian work¹³ claims to show that the rate of atmospheric corrosion drops sharply at temperatures below -25°C approaching zero at -45°C . The mean temperature at Norman Wells is above -25°C for 9 months of the year and above zero for 5 months. Precipitation is considerable throughout the year and heaviest in the summer months. The relative humidity is consistently greater than 85% and it has been proposed that in general the 'time of wetness' of a corroding surface is that for which the relative humidity exceeds 85%. The authors would be interested to receive comments on these points or details of

Table 3. Periods of exposure and corrosion rates of low-alloy copper steels in Antarctica

Exposure site	Steel type	Date exposed	Date recovered	Time of exposure (days)	Corrosion rate ($\mu\text{m/y}$)
Vanda Station	BHP	30 Dec 83	24 Dec 84	360	0.87
Scott's Hut,	BHP	30 Dec 83(a)	29 Dec 84	365	10.83
Cape Evans	BISRA	30 Dec 83(a)	29 Dec 84	365	10.20
"	BHP	30 Dec 83(a)	17 Nov 86	1053	4.42

(a) Approximate date of exposure

Table 4. Corrosion rates of carbon steel and ingot iron in subpolar and hot arid climates

Exposure Site	Country	Latitude	Longitude	Climate type	Material (see footnote)	Corrosion Rate (μm/y) Exposure period (years)			Reference
						1	2	3	
Norman Wells	Canada	65°15'N	127°38'W	Cold-Continental	Cold rolled steel (a)	0.7	1.2		10
Abisko	Sweden	68°30'N	19°E		Sub-polar	Ingot iron (b)	5.1		
Phoenix	USA	33°28'N	112°04'W	Desert	Cold rolled steel (a)	6.3	4.4		10
Khartoum	The Sudan	15°37'N	32°33'E	Semi-arid	Ingot iron (b)	2.5			11
Mt Newman	Australia	23°S	119°30'E	Semi-arid	Plain carbon steel (c)	3.0	2.0	1.0	8

- (a) Initially cleaned in acid.
(b) Specimens cut from hot-rolled material and initially cleaned in hydrochloric acid.
(c) Panels exposed in the 'as rolled' and abrasive blast cleaned condition.

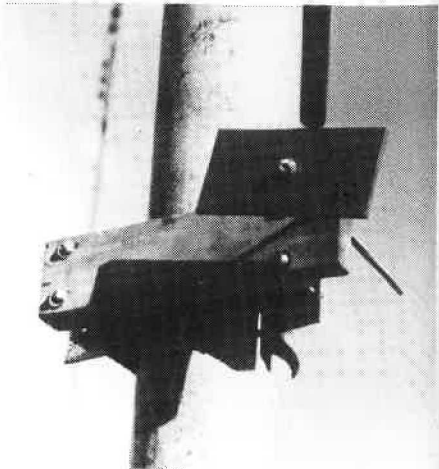
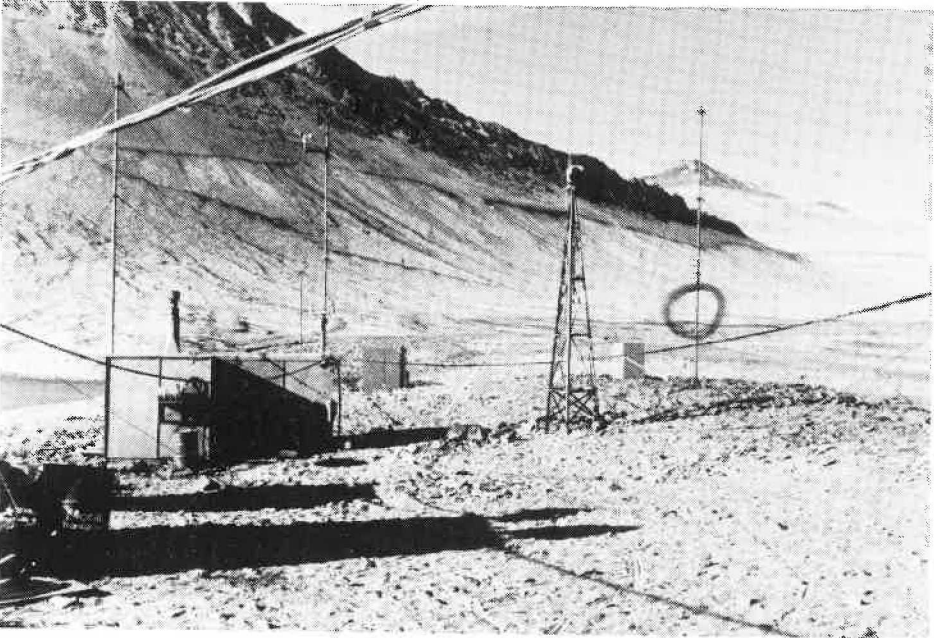


Figure 3. Detail of mounting of corrosion specimen on antenna mast.

Figure 2. Inland site at Vanda Station in the 'dry valleys'. Specimen located on radio antenna mast as indicated.

any recent measurements made in the Arctic.

4. CONCLUSIONS

It has been shown that external atmospheric corrosion at coastal sites in Antarctica is a sufficient problem to require the use of moderate levels of corrosion protection as normally employed externally in temperate climate zones. It follows that the clearing of ice and snow from historic huts and so changing the microenvironment must lead to the continuing deterioration of historic metallic components and contents.

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Table 5. Climatic Parameters for Norman Wells North West Territory Canada (elevation 88m) (see ref. 12)

	Mean monthly temperature °C	Precipitation water equivalent (mm)	Relative Humidity % 0400 hrs	1000 hrs
January	-28	18	90	88
February	-27	15	85	81
March	-18	8	85	84
April	- 8	13	85	71
May	+ 6	18	81	65
June	+14	36	77	59
July	+16	51	81	63
August	+13	69	88	71
September	+ 6	43	89	81
October	- 4	20	90	87
November	-17	20	94	94
December	-25	18	90	89
Annual	- 6		86	78
Total		329		

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